

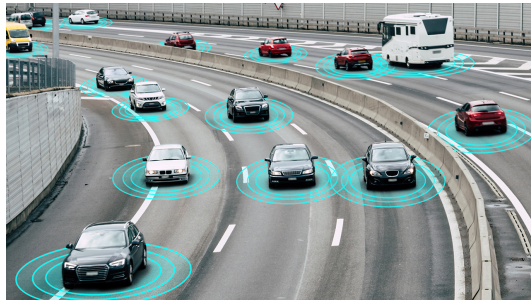
High-Mobility Wireless Channel Measurements at 5.9 GHz in an Urban Environment

F. Pašić, S. Pratschner, R. Langwieser and Christoph F. Mecklenbräuker

Presenter: Faruk Pašić

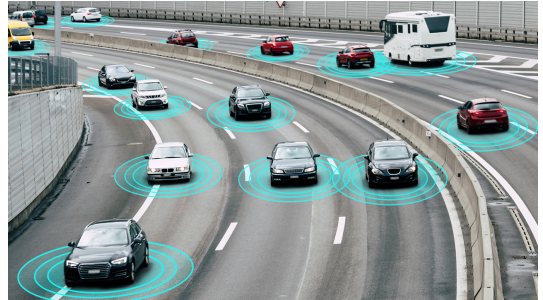
Introduction

- demand for transportation rapidly increased



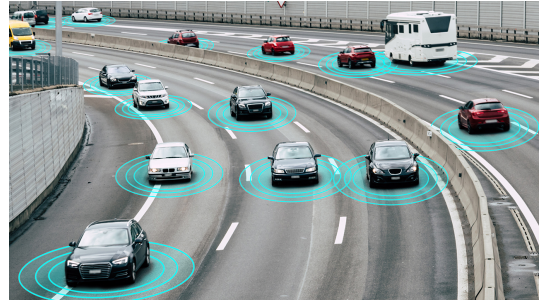
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 - traffic accidents
 - congestion



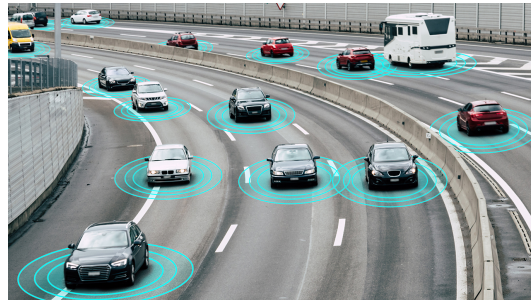
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- use vehicular applications
 - improve transportation quality and safety



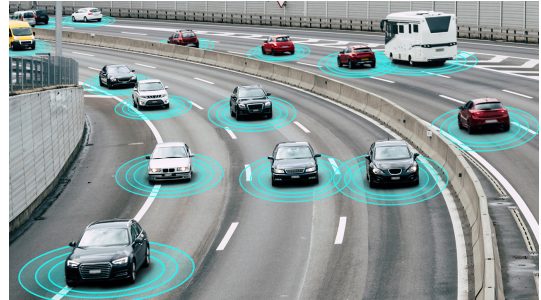
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 - require high reliability and low latency in high-mobility scenarios



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 - require high reliability and low latency in high-mobility scenarios
 - unlicensed ITS band at 5.9 GHz



Motivation

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 - control environmental parameters for each measurement repetition
- result: different small-scale fading behavior

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 - same measured path for different transmitter velocities
 - different times, but without losing repeatability
 - testbed hardware for repeatable and controllable high-speed measurements
- **results of the measured wireless channel compared in terms of delay-Doppler function**

Testbed Architecture

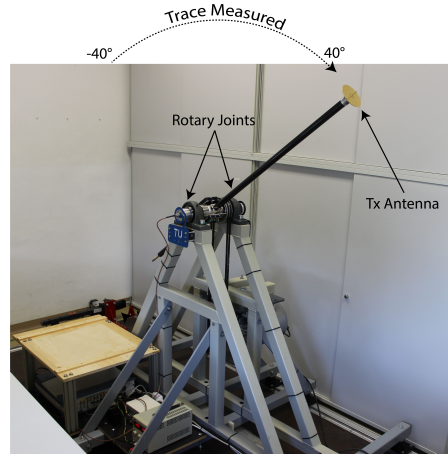
Measurement Campaign

Measurement Results

Summary and Conclusion

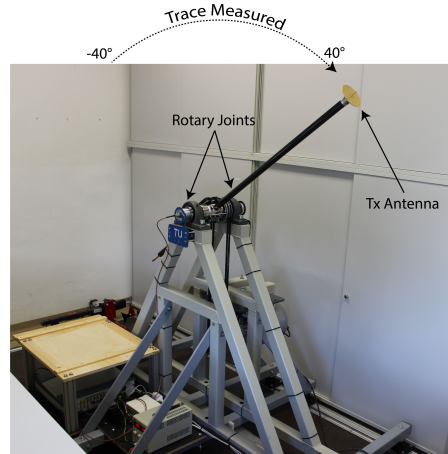
Testbed Architecture - General Description

- rotary unit
 - trigger unit
 - rotary joints
 - repeatable measurements up to 400 km/h in controlled environment



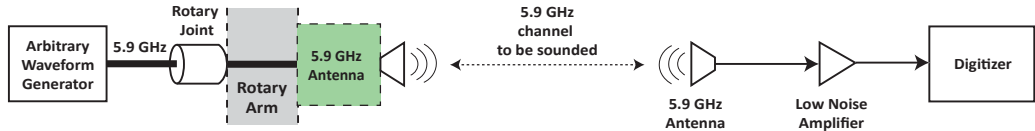
Testbed Architecture - General Description

- rotary unit
 - trigger unit
 - rotary joints
 - repeatable measurements up to 400 km/h in controlled environment
- idea for repeatable channel sounding measurements at 5.9 GHz
 - rotating Tx and static Rx
 - perform channel measurements along the same trace for different velocities



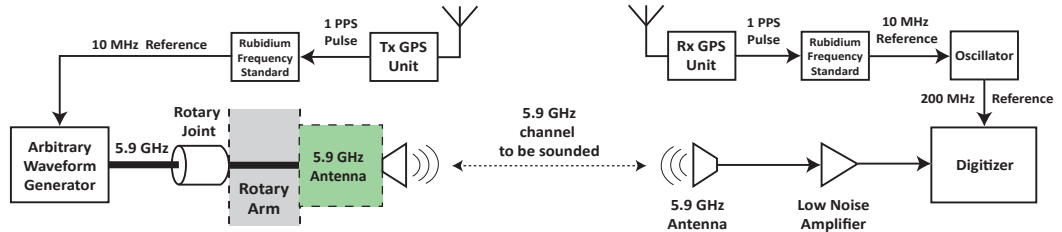
Testbed Architecture - Details

- Tx signal transmitted at a center frequency $f_{RF} = 5.9$ GHz



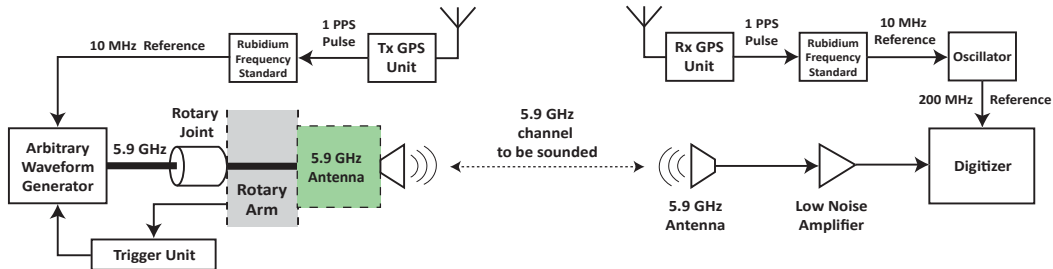
Testbed Architecture - Details

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- frequency synchronization: GPS Unit and Rubidium Frequency Standard



Testbed Architecture - Details

- Tx signal transmitted at a center frequency $f_{RF} = 5.9$ GHz
- frequency synchronization: GPS Unit and Rubidium Frequency Standard
- time synchronization: autocorrelation properties of Zadoff-Chu sequence



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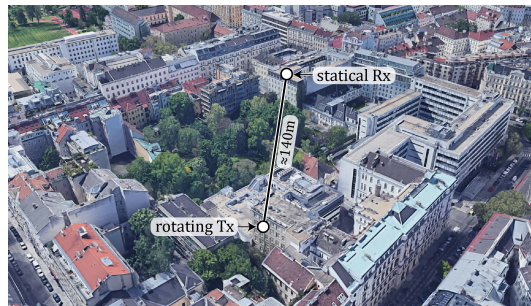
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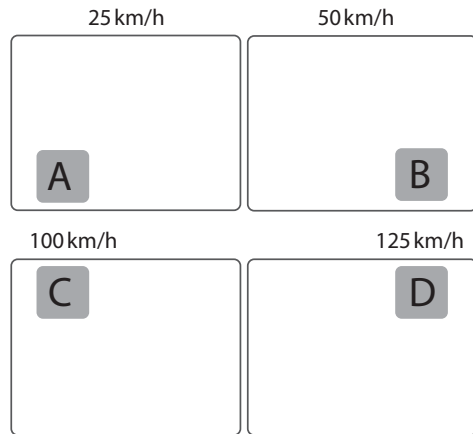
Measurement Campaign

- wireless channel measurements
 - moving Tx antenna
 - static Rx antenna



Measurement Campaign

- wireless channel measurements
 - moving Tx antenna
 - statical Rx antenna
- measurement procedure
 - channel sounding using OFDM



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 - low velocity $\rightarrow$ low subcarrier spacing

25 km/h

subcarrier spacing = 250 kHz
subcarriers = 400
symbol duration = 4 μ s
meas. duration = 200 ms
max. Doppler shift = 136 Hz

A

50 km/h

subcarrier spacing = 500 kHz
subcarriers = 200
symbol duration = 2 μ s
meas. duration = 100 ms
max. Doppler shift = 273 Hz

B

100 km/h

C

125 km/h

D

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meas. duration = 100 ms
max. Doppler shift = 273 Hz

B

100 km/h

subcarrier spacing = 1 MHz
subcarriers = 100
symbol duration = 1 μ s
meas. duration = 50 ms
max. Doppler shift = 546 Hz

C

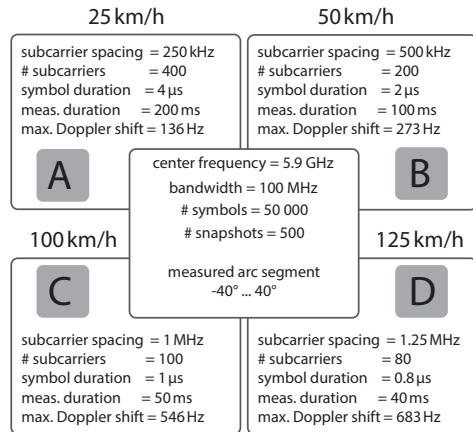
125 km/h

subcarrier spacing = 1.25 MHz
subcarriers = 80
symbol duration = 0.8 μ s
meas. duration = 40 ms
max. Doppler shift = 683 Hz

D

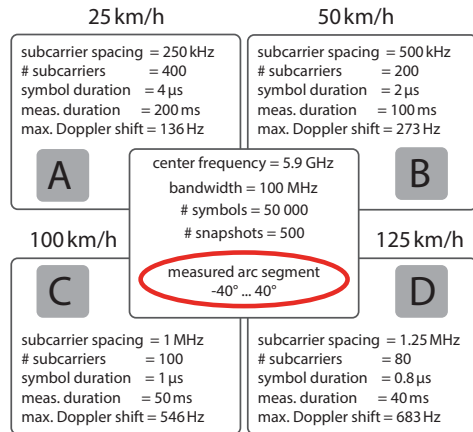
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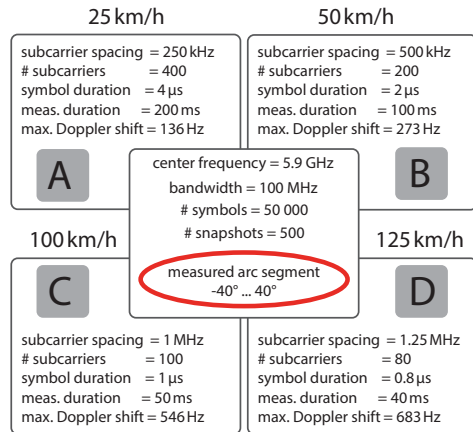
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 - low velocity $\rightarrow$ low subcarrier spacing
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 - comparability of results achieved



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- wireless channel measurements
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 - comparability of results achieved
- result: time-variant channel transfer function



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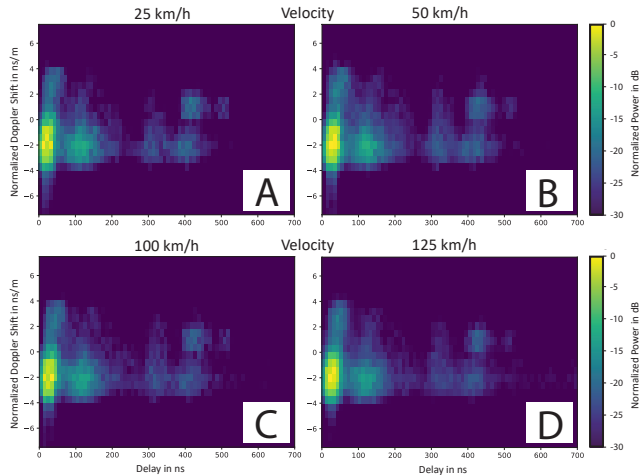
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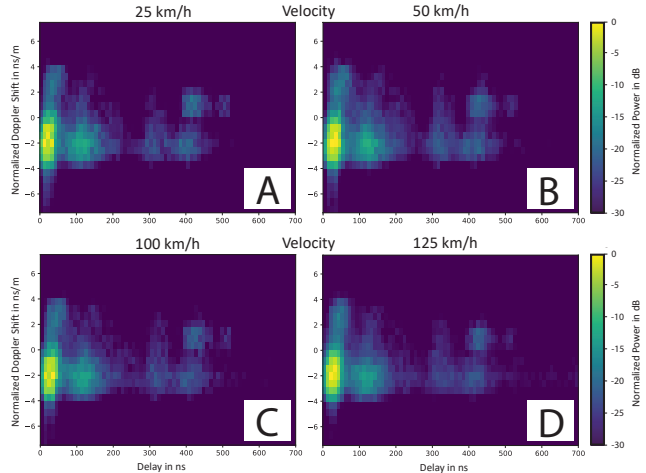
Measurement Results

- results in terms of delay-Doppler functions



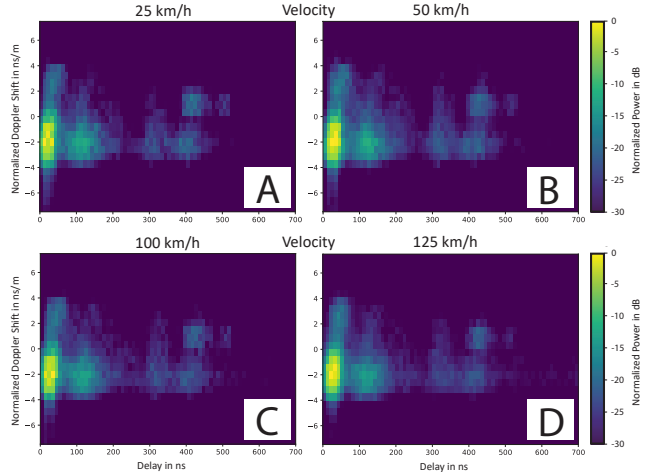
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- results in terms of delay-Doppler functions
- normalizations performed:
 - normalized Doppler Shift
 - intensity normalized to 0 dB



Measurement Results

- results in terms of delay-Doppler functions
- normalizations performed:
 - normalized Doppler Shift
 - intensity normalized to 0 dB
- key observations for delay-Doppler functions:
 - invariance with regard to the transmitter's velocity
 - the main Doppler component is negative



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- **repeatable measurements in high-mobility urban environment are possible**

Summary and Conclusion

- **repeatable measurements in high-mobility urban environment are possible**
- **proposed channel sounding system works correctly**

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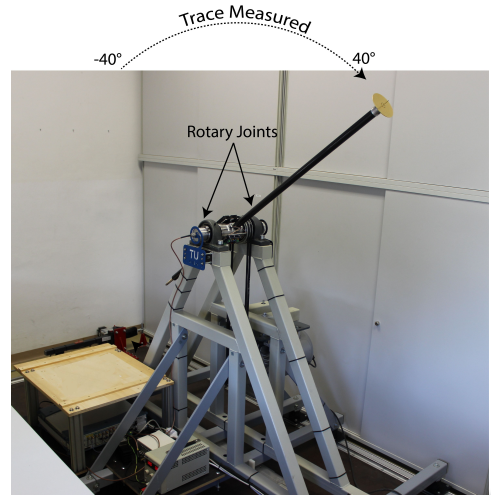


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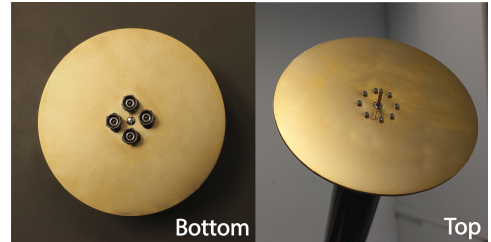
Why Rotating Tx and Static Rx ?

- rotary unit can be used either as Tx or Rx
- employing rotary unit as Rx leads to losing the time synchronization



Antennas

- rotating Tx antenna
 - self-built and lightweight
 - vertical dipole
 - $\frac{\lambda}{4}$ brass pipe soldered on the PCB
 - attached to rotary arm with 4 connectors



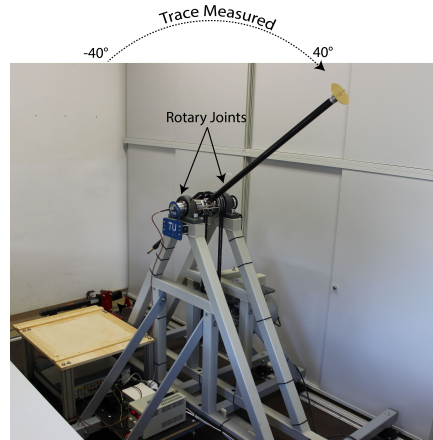
Antennas

- rotating Tx antenna
 - self-built and lightweight
 - vertical dipole
 - $\frac{\lambda}{4}$ brass pipe soldered on the PCB
 - attached to rotary arm with 4 connectors
- static Rx antenna
 - multiple dipole antenna
 - 4 elements stacked in vertical direction
 - omnidirectional pattern in the horizontal plane



Rotary Unit

- repeatable, closed-loop, off-line-processed measurements at very high velocities
 - 1500 revolutions per minute (up to 400 km/h)
 - aluminum alloy arm (length of 1m)
 - semi-rigid coaxial radio-frequency cables
- repeatability
 - keep the surroundings constant
 - no objects move
 - repeat a position with accuracy of 100 μm
- controllability
 - change average SNR (Tx power)
 - change velocity (rotation speed)
- measurement repetition rate
 - realistic vehicular scenario $\rightarrow$ few measurements until the vehicle has passed the transmitter
 - many measurements in short time-spans



Trigger Unit

- used for time synchronization
- uses rotational encoder
- trigger measurement at precisely defined position of rotary arm
- 960 possible angle positions (accuracy of $300\text{ }\mu\text{m}$)
- signal of rotary encoder decoded by counter and comparator to form trigger signal
- trigger signal fed to Tx-Generator

Rubidium Frequency Standard

- SRS PRS10 Rubidium Frequency Standard
- Standard output: 0.5 Vrms 10 MHz sine wave
- maximum frequency deviation: $\pm 5 \cdot 10^{-11}$
- phase noise < 130 dBc/Hz
- receives 1 pps input from GPS Unit

Local Scattering Function

- non-stationary time-variant frequency transfer function obtained as result
- assumption: channel is locally stationary
 - 6λ of motion (100 snapshots for 5.9 GHz case)
 - entire frequency range
- calculate local scattering function according to [3] and [4]:
 - group non-stationary measured frequency transfer function into more local stationary frequency transfer functions
 - for each of them perform FFT to obtain local stationary delay-Doppler functions
 - average delay-Doppler functions over number of taken local stationary groups

[3] M. Hofer et al., “Wireless vehicular multiband measurements in centimeterwave and millimeterwave bands” in 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Helsinki, Finland, Sep. 2021.

[4] L. Bernado et al., “Delay and Doppler spreads of nonstationary vehicular channels for safety-relevant scenarios”, IEEE Transactions on Vehicular Technology, vol. 63, no. 1, 2014.